Kexuan (Michael) Huang

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Graduate Student

EDUCATION

University of Michigan 2022.08
- M.S. in Information Ann Arbor, MI, U.S.

Shanghai Jiao Tong University 2018.09
- B.S.E in Electrical and Computer Engineering Shanghai, China
- Minor in Entrepreneurship

University of Wisconsin-Madison 2022.01 - 2022.05
- Exchange Student in Computer Science Madison, WI, U.S.

INTERNSHIPS

Advanced Micro Devices, Inc. (AMD) 2020.12 - 2021.05
Design Verification Engineer Intern, dGPU SOC, Shanghai R&D Center Shanghai, China

- Work on *HDL* code and waveform simulation to do RTL troubleshooting.
- Use *Python*, *Shell*, *Perl* & *Ruby* to code automation scripts and maintain regression flow. Use *Pandas*, *NumPy*, and *Sklearn* to analyze data, use *Requests and RegEx* to scrape website, and use *Django* to set up team webpage.

Segway-Ninebot, Inc. 2020.01 - 2020.03
Embedded Software Engineer Intern, IoT R&D Department. Shanghai, China

- Work on development of autonomous delivery robot, typically in communication & decision-making strategy optimization.
- Use *C* to do embedded programming with *STM* chip, prototype PCB verification and feedback reporting.

UM-SJTU Joint Institute 2021.05 - 2021.12
Teaching Assistant Shanghai, China

- Hold workshops and Q&A sessions, grade assignments & exams to help nearly 100 students in courses VE300 & VE496.

UM-SJTU Joint Institute 2021.05 - 2021.08
Research Intern Shanghai, China

- Reproduce, simulate and optimize high-speed signal-free intersection models in *SUMO* software with *Python*.

PROJECTS

Intelligent firefighting system based on Internet-of-Things technology, Frontend Team Leader 2021.07 - 2021.12
Capstone Project, UM-SJTU Joint Institute

- Develop an integrated IoT firefighting system to perform fire risk management with greater accuracy, better efficiency and lower cost using *FAHP* algorithm. The system is deployed on servers of Shanghai fire brigade currently.
- Implement backend with *FastAPI,* webpage with *React.js* and a cross-platform app with *React Native*. Use MongoDB as database, whose current data-handling quantity is at 7 million.

Digital Innovation of the Fever Thermometer, Team Leader 2020.09 - 2021.03
PRP Project (Excellent Project: 3 in 26), Antai College of Economics & Management

- Develop a *WeChat* mini-program with *JavaScript, HTML* and *CSS.* Design the circuit and PCB of infrared thermometer, which can connect to mobiles via *Bluetooth*. Temperature registration can be done with greater efficiency in COVID-19 pandemic.

The Noisy Town, Developer 2021.09 - 2021.12
Tencent Game Development Project (Silver Award)

- Use *Roblox* Game Engine to develop an open world, task-driven, role-playing simulation game, especially for children around 6~12 years old to explore professional interests through game-based learning.

HONORS

2020-2021 Merit Undergraduate Scholarship, SJTU 2021.11
2019-2020 Most Improved Scholarship, SJTU 2020.10
Advanced Teaching Assistant, UM-SJTU Joint Institute 2021.08

SKILLS

- **Languages:** English (IELTS 7.5 / CET-6), Chinese (native)
- **Programming:** Assembly, Verilog, C (Linux), C++, Python, MATLAB, TypeScript, HTML, Shell, SQL, Markdown, LaTeX
- **Development Tools:** Git, GitHub, Drone CI, Vim, VS Code, macOS, Linux
- **Hobbies:** Badminton (former professional athlete, former SJTU badminton team member), Anime Piano